pfi

Python for Fixed Income

COURSE 5 · ADVANCED MACHINE LEARNING

Random Forests

WEEK 1 · DAY 3

Created by **Jeff Lenamon**

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Trees that agree do not average away much

The variance of an average of B numbers with correlation rho

```
rho x sigma^2   +   (1 - rho) x sigma^2 / B
```

rho = 1

Trees agree perfectly: averaging removes nothing, however many there are.

rho = 0

Trees share nothing: a large enough average removes all of their noise.

0.2581

tree correlation, bagging

100 of 100

trees asking PAY_0 first

How tree correlation works

The problem	Averaging helps only as far as the trees disagree, so measure how much they agree
Score	Every tree scores the validation accounts: one column of outputs per tree
Correlate	Correlate every pair of columns and average the pairs, each pair once
Read it	Near 1: the trees say the same thing. Near 0: each adds something the others lack
What comes out	One number: the average pairwise correlation of the trees' outputs

How a random forest works

The problem	Bagged trees all ask about the strongest column first, so they err alike
Rows	Each tree is fitted on its own bootstrap sample of the rows
Columns	At every split a tree may choose only among a random subset of the columns
Grow and average	Grow each tree deep, then average the trees' class-1 probabilities
You control	n_estimators, max_features (columns per split), and min_samples_leaf

A random subset of columns at every split

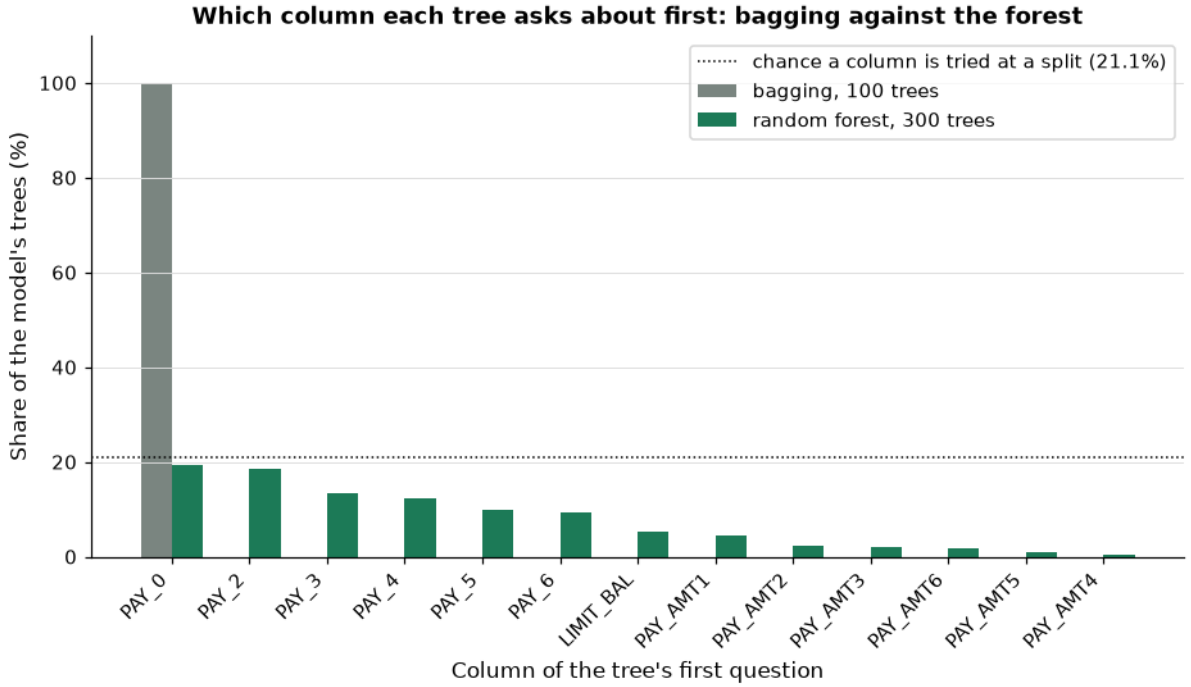
4 of 19

columns tried at each split ("sqrt")

0.2105

chance a given column is tried

PAY_0 comes first in 58 of 300 trees (19.3 percent), about as often as it is in the root's draw at all. The first questions use 13 different columns; bagging's 100 trees all start with PAY_0.



A column a split cannot see cannot be chosen there: the trees differ from the first question.

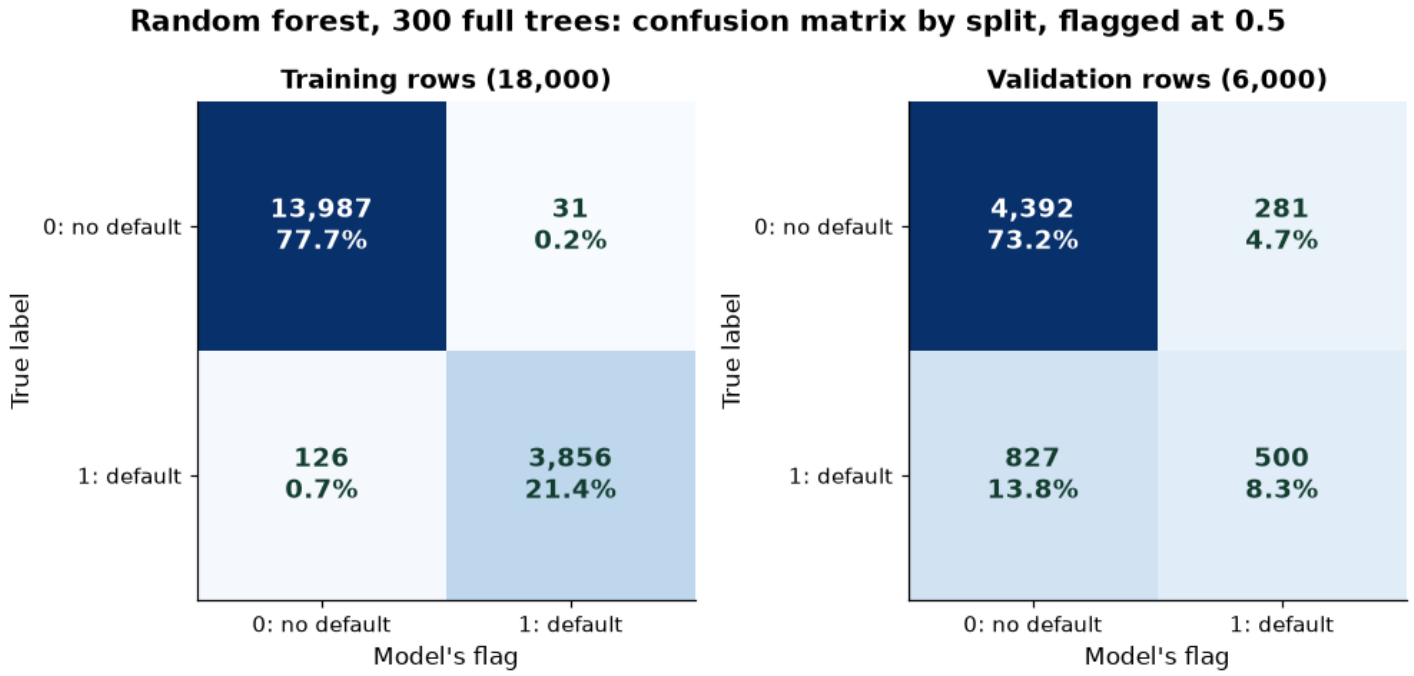
The forest beside the bagged model: full trees

Full trees	Validation ROC AUC	Average precision	Tree correlation
Bagging, 100	0.7602	0.5105	0.2581
Random forest, 300	0.7600	0.5146	0.2487

With full trees, the random columns change the first questions and barely change the agreement or the ranking.

Full trees already disagree on many accounts, so there is little agreement left to remove.

The standard view: one heatmap per split



Training, then validation, on every card day. The test rows never get a panel.

How to read it

True label down the side, the model's flag at 0.5 across the top. Each cell: a count and its share of that split's rows.

Full trees

Training panel: almost all on the diagonal. Validation panel: far from it (ROC AUC 0.7600). Bagging's panels look the same.

The name on today's class

Leo Breiman, October 2001

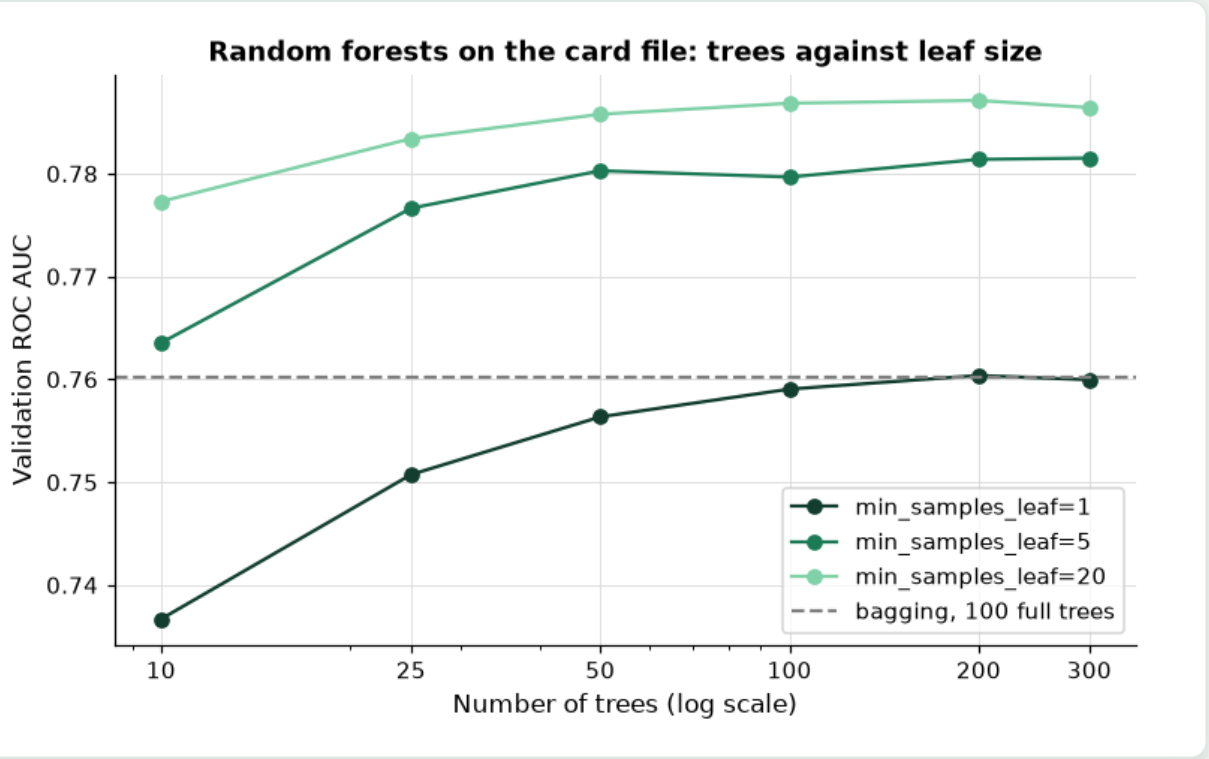
"Random Forests"

Machine Learning, volume 45, issue 1, pages 5 to 32

The same author as week 1's 1996 paper; the title is scikit-learn's class name.

Source, read 5 October 2026: the Crossref record of the paper (DOI 10.1023/A:1010933404324). Bibliographic facts only; the paper itself was not read for this course.

Leaf size matters more than tree count here

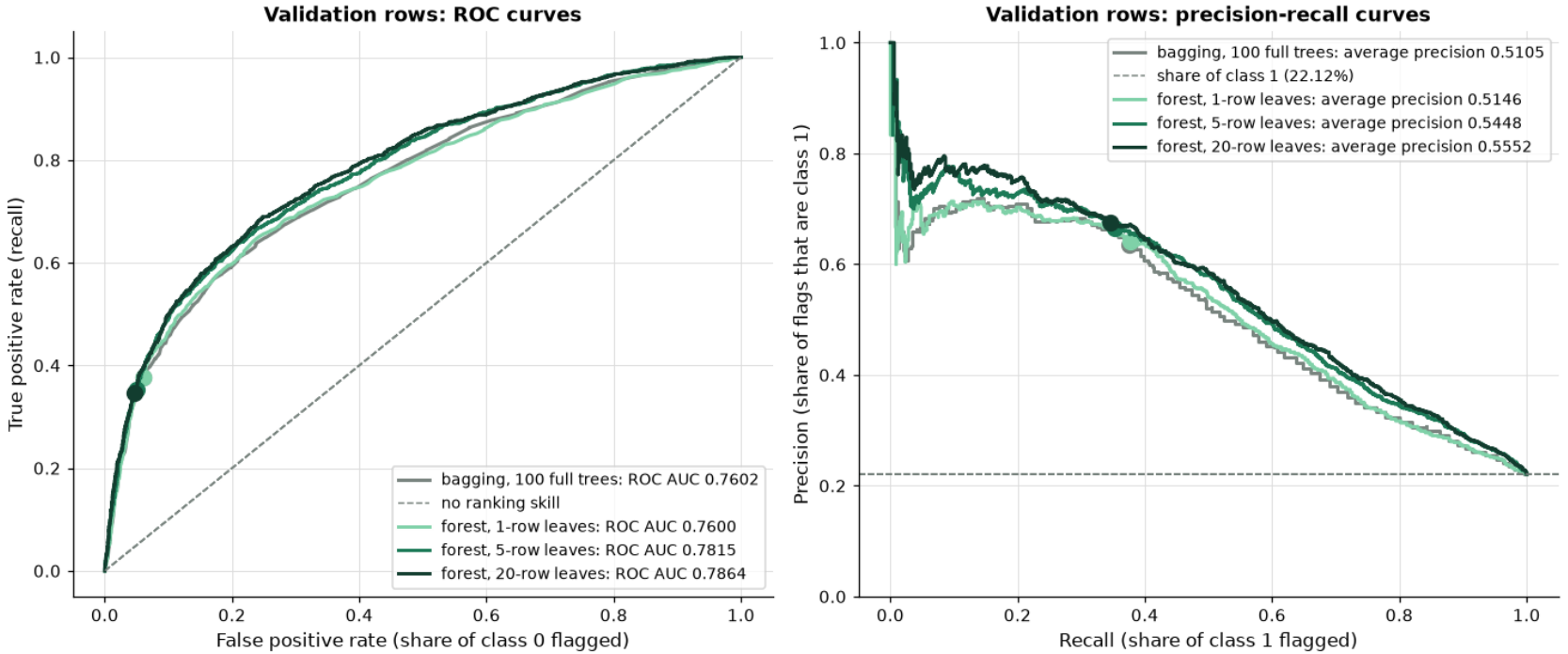


Min leaf	One tree	Correlation	300 trees
1	0.6144	0.2487	0.7600
5	0.6801	0.4116	0.7815
20	0.7257	0.6311	0.7864

Bigger leaves: the trees agree more, each tree alone is much better, and the forest improves.

An average is as good as its members and as independent as they are. On this file, better members win.

The curves by leaf size: validation rows



Bigger leaves: higher curves, and the dot at 0.5 moves left as fewer accounts reach it.

max_features=None is bagging

100 full trees	Columns per split	Validation ROC AUC	Tree correlation
RandomForestClassifier, max_features=None	19	0.7595	0.2577
BaggingClassifier	19	0.7602	0.2581

Equal settings, nearly equal numbers: every tree asks PAY_0 first. Today's AI check.

How class weights work

The problem	Defaults are a small share, so impurity splits attend mostly to the other accounts
Weight	A class weight counts each row of a class as more or less than one row
balanced	Each class gets weight n / (2 x its count); balanced_subsample redoes it per bootstrap sample
You control	The choice of weights
What comes out	A similar ranking at a higher level: the model's score, not a probability of default

Class weights: the same ranking, a higher level

2.2602

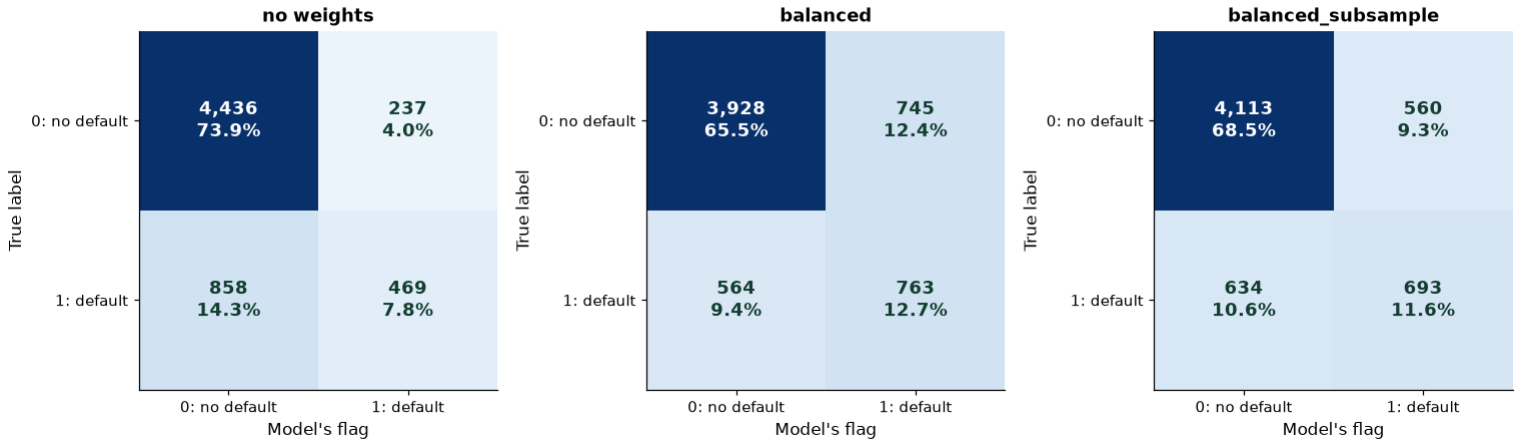
weight of each default

0.7806

ROC AUC, balanced

0.3735

mean output, balanced

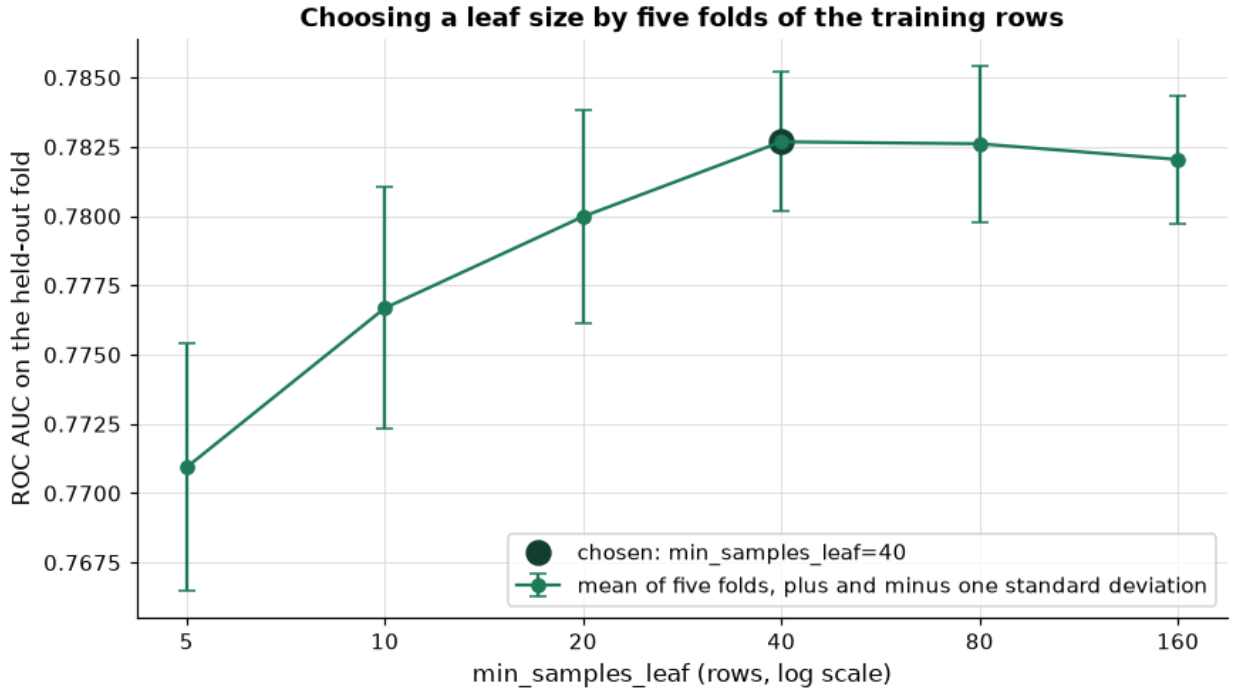


Unweighted: ROC AUC 0.7815, mean output 0.2223, default rate 0.2212. A weighted forest's output is the model's score.

How a cross-validated search works

The problem	A setting cannot be chosen on rows the forest was fitted on, or on the validation rows
Folds	Cut the training rows into five; fit on four, score on the fifth, five times
Every candidate	GridSearchCV does this for each leaf size and averages the five scores
You control	The candidates, the seeded folds, and the metric (ROC AUC)
What comes out	Mean fold scores, the best leaf size, and a forest refitted on all the training rows

Choosing a leaf size on the training rows



40

rows per leaf, chosen

0.7844

validation ROC AUC, reported once

50-tree forests, five folds of the training rows. Fold mean of the chosen leaf size: 0.7827, the best of six noisy numbers.

The validation rows look and report. They never choose.

How extra trees work

The problem	A forest still searches every threshold, and that search is where a tree fits its sample closely
Random thresholds	Draw a few columns, draw one threshold per column at random, keep the best random question
Rows	By default each tree sees all the training rows: no bootstrap
Trade	Each tree is a little weaker than a forest's: its thresholds are not the best ones
You control	The same settings as a forest; the output is again an average

Every setting of RandomForestClassifier

Setting	Default	Tune first?
n_estimators	100	
max_depth	None	
min_samples_leaf	1	tune first
max_features	'sqrt'	tune first
bootstrap	True	
max_samples	None	tune first
class_weight	None	
oob_score	False	
criterion	'gini'	
ccp_alpha	0.0	

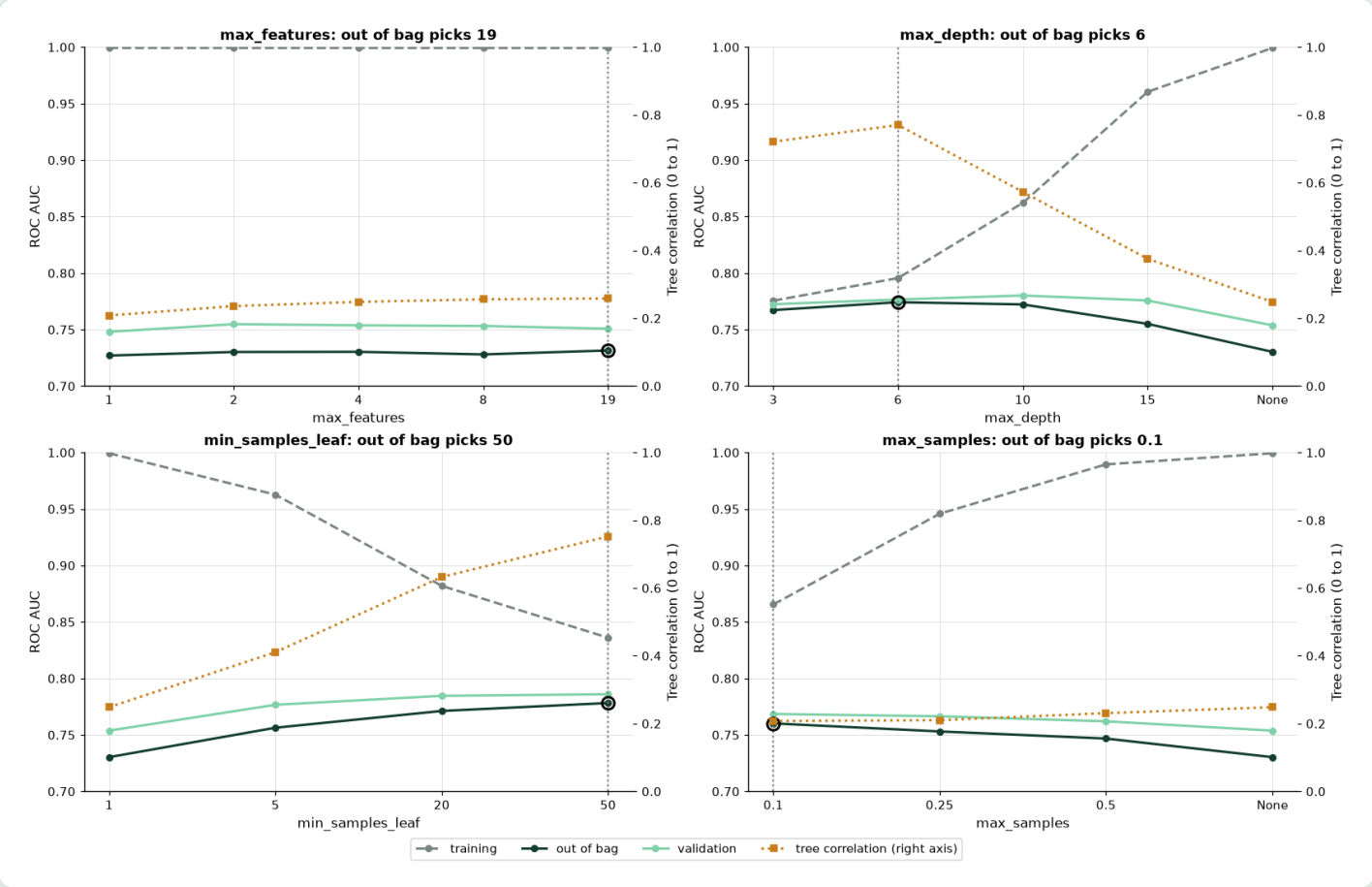
Nineteen settings, by group

Tree shape (max_depth, min_samples_leaf), randomness (max_features, bootstrap, max_samples, seed), regularization, class weights, and housekeeping. Ten of the nineteen are shown.

The three to tune first

min_samples_leaf, max_features, and max_samples: they decide how strong the trees are and how much they agree. max_depth does the leaf-size job from the other side.

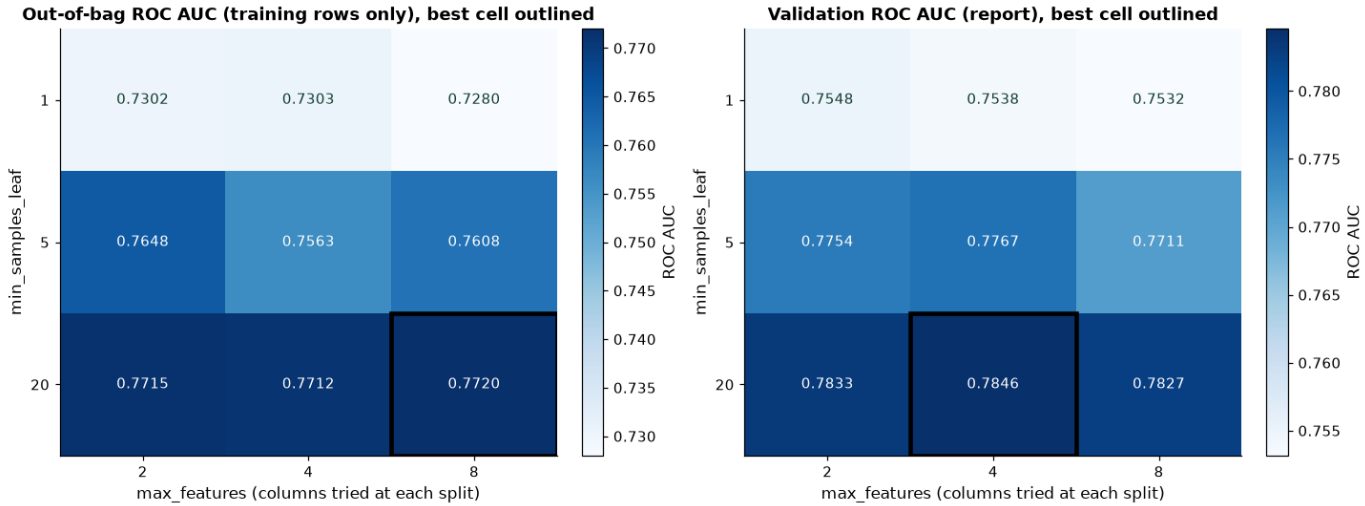
Four settings, one at a time, chosen out of bag



Setting	Pick	OOB	Valid.	Corr.
max_features	19	0.7315	0.7508	0.2584
max_depth	6	0.7742	0.7766	0.7703
min_samples_leaf	50	0.7782	0.7860	0.7523
max_samples	0.1	0.7602	0.7685	0.2078

Dotted orange: how much the trees agree. min_samples_leaf moves the ranking; max_features moves the agreement.

Two settings together, and extra trees



Extra trees, 100	ROC AUC	AP
bootstrap=False (default)	0.7792	0.5473
bootstrap=True	0.7818	0.5509

One default differs

ExtraTreesClassifier takes a forest's settings. Only bootstrap differs. The random thresholds are not a setting.

Leaf size is the setting to tune first on this file; columns per split stays at its default unless there is a reason.

Excel side by side

Excel side by side (tested in Excel)

`=INT(SQRT(19))`	4 columns per split, "sqrt"
`=INT(0.5*19)`	9 columns per split, max_features=0.5
`=1-COMBIN(18,4)/COMBIN(19,4)`	0.2105, a given column tried
`=CORREL(B2:B6001,C2:C6001)`	bagging 0.2564, forest 0.2362
`ten CORREL pairs, then =AVERAGE(H1:H10)`	0.2455, five trees
`=ROWS(A2:A18001)/(2*COUNTIF(A2:A18001,1))`	2.2602, weight of a default
`=AVERAGE(C1:C5) and =STDEV.P(C1:C5)`	the search's mean and SD

No forest and no tree in Excel. The arithmetic around one, typed into Excel by script and checked against Python.

Save the day: one file's history

```
git log --oneline -- results.csv
```

```
Experiment log: leaf sizes, class weights, the searched forest, extra trees
Experiment log: bagging and a random forest of full trees, validation rows
```

```
model,rows,roc_auc,average_precision,threshold,accuracy,precision,recall,f1
bagging_100_full_trees,validation,0.7602,0.5105,0.5000,0.8143,0.6357,0.3760,0.4725
random_forest_300_full_trees,validation,0.7600,0.5146,0.5000,0.8153,0.6402,0.3768,0.4744
random_forest_300_leaf_5,validation,0.7815,0.5448,0.5000,0.8175,0.6643,0.3534,0.4614
random_forest_300_leaf_20,validation,0.7864,0.5552,0.5000,0.8185,0.6750,0.3459,0.4574
random_forest_300_leaf_5_balanced,validation,0.7806,0.5386,0.5000,0.7818,0.5060,0.5750,0.5383
random_forest_50_leaf_40_searched,validation,0.7844,0.5534,0.5000,0.8198,0.6814,0.3482,0.4608
extra_trees_300_leaf_5,validation,0.7814,0.5490,0.5000,0.8202,0.6782,0.3557,0.4666
```

Two commits touched results.csv; the code commit did not. A path after -- gives that file's history alone.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

Docstring

Which rows fit, which rows choose, which rows report: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: a "random forest" with max_features=None: bagging under another class name.

Code that runs is not the same as code that is right.

What you can do now

- Measure how much an ensemble's trees agree
- Work out the columns a split tries
- max_features=None is bagging
- Leaf size moves a forest more than tree count
- Class weights: same ranking, the model's score

Week 1, Day 4 Impurity and permutation importance